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國立臺灣大學期中學期考試答案卷

National Taiwan University Midterm/Final Examination Answer Sheet

課程編號 Course no.

科目 Course title 作業系統 　　 管理 學院 College 資訊管理 學系 Department 　 組 Division 二 年級 Year

考試日期 Date 2016 年 4 月 19 日 　 學號 Student ID no. b03705002 　 姓名 Name 林軒逸

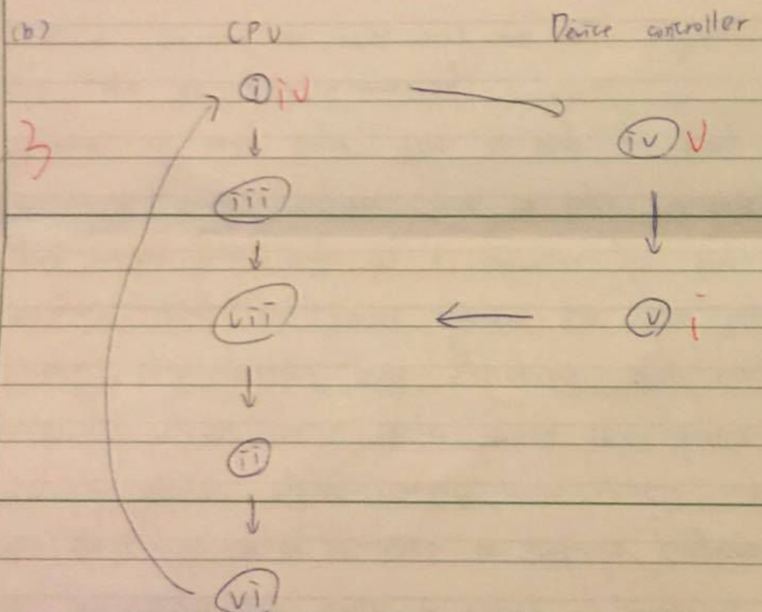
從此處開始寫起，試卷用紙務須節用，非經主試認可不得擅用其他紙張作答。

記分欄 Please write from here.

1.

(a) Modern interrupt mechanism accelerates interrupt process by interrupt vector. The interrupt sent by I/O devices will contain a index of the interrupt (in register), which represents what it wants to do, and the interrupt handler will check the index in the interrupt vector, performing a quick response and return to the interrupted process, in a quick response time. Also, it doesn't have to keep checking I/O status, since I/O devices will notify interrupt handler by interrupt request line when it's ready.

(b) -3



(c) The benefit of multiprogramming is the CPU never idles. That is, we can make max utility of CPU. Multiprogramming is done by job scheduling and selecting, ensuring that there must be available job for CPU to execute at any moment. Time sharing is an extension of multiprogramming, and keeps switching between jobs and lower the response time. To differ multiprogramming and time sharing, multiprogramming makes the max CPU use, and time sharing is to minimize response time.

(d) Operating systems support new devices by device drivers. Device drivers are softwares, as medium of device controllers and communicate with operating system. In this way, devices can be recognized by any operating systems and be supported.

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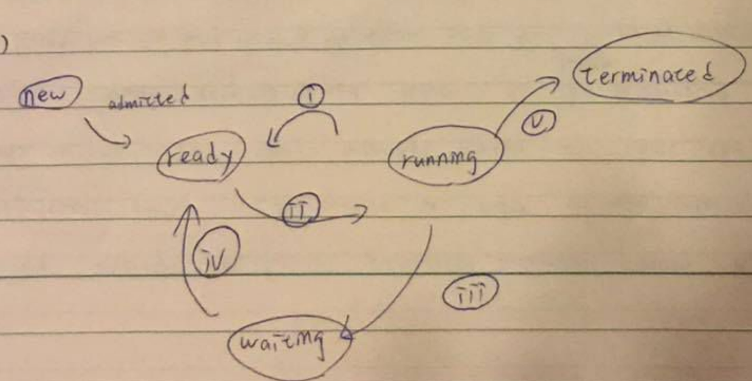
2. (a) When user uses an API function, such as read(), the API will do the system call, the software interrupt for user. The API function will load the index of (system call number) the system call to register, and change the mode from user mode to after software interrupt kernel mode. After kernel received the system call, the kernel will check the table of code pointer, knowing what the system call wants to do and return the data or do the specific routine. After that, it returns to user mode and continue executing.

(b) Since directly doing system calls is hard, and changing modes is dangerous and difficult to implement, Thus, programmers prefer to use API functions to do system call, which is easier to understand and implement, also safer to do. Also, the program can be recognized on different computers or operating systems.

(c) layer operating system structure is easy for debugging, since the bug in each layer must be in the specific layer. Also, it protects the hardware from being directly harmed, thus provides more protections than simple structures. However, defining layers isn't a easy job, what function does this layer have and how to divide it with another layer might be a difficult job. Also, in layer structure, system calls might be tangled together. Since in layer structure, only high level layers can use functions of lower layer. Thus, it might do many system calls from layer 0 → N and N → n and might be time-consuming.

3. (a)



-2

(b) value = 10

value = 250

global variables 彼此不 share

If we remove the wait() statement, we can't ensure we can have the child function finished first, and the parent and child will run concurrently, not ensuring who will get the CPU runs first. Thus, the output is unpredictable.

output = 20 10

the child function finished first, and the parent ...
not ensuring who will get the CPU runs first. Thus the output is unpredictable.

output = 20 10 or 10 20

(看scheduler怎麼安排)

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4.

share memory

(a) Since threads use some resources together, such as registers, creating a thread is easier to implement and less time-consuming since what needs to be created is less. However, creating a process needs to create text-segment, data-segment, stack, heap at the same time, which takes more time to implement, and also much more costly.

(b) One-to-one thread model can be run on multiprocessors. That is, supporting parallel execution. Also, if a thread is blocked due to system call, other threads can still keep executing. However, one-to-one thread model has a restriction on thread amount. There can't be too many threads and might be unable to efficiently run a program, or even can't run a program.

5.

```
int main (void)
{
  char inputBuffer [80];
  bool background;
  while (cin)   // while user is inputting, and haven't close the program.
  {
    cout << endl << " COMMAND -> ";
    pid_t pid;
    cin >> inputBuffer;
    for (int i=0, i<79, i++)
    {
      if (inputBuffer[i] == '&' & inputBuffer[i+1] ==NULL) // the last character of the buffer
      {
        background = true;
        break;
      }
    }
    if (char. inputBuffer [79]. == '&')
        background = true;
```

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```
pid = fork();
if (pid == 0)
    exec (inputBuffer);
else if (pid > 0) {
    if (background == false)
        wait (NULL);
    exit(0);
  }
 } // end of while
} // end of main
```

pid < 0 case ;

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